

Amit Adate

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Research Interests

Computer Vision, Deep Learning, Machine Learning, Optimization

Education

- 2020 – 2024 **Northwestern University, USA.**
Ph.D. in Electrical Engineering
Advisor: Dr. Aggelos K Katsaggelos
- 2018 – 2019 **Northwestern University, USA.**
Masters of Science in Computer Science with Specialization in Artificial Intelligence
Advisor: Dr. Aggelos K Katsaggelos
- 2014 – 2018 **Vellore Institute of Technology, India.**
Bachelor of Technology in Computer Science
Advisor: Dr. B. K. Tripathy

Professional Experience

- January 2019 **Graduate Research Assistant, Northwestern University.**
– Present Image and Video Processing Lab
Guide: Dr Aggelos K Katsaggelos
- August 2017 **Undergraduate Research Assistant, VIT University.**
– July 2018 Soft Computing Lab
Guide: Dr B K Tripathy
- May 2017 **Research Intern, CISCO Systems, Inc.**
– July 2017 Advanced Services Team
Worked on Large Scale Parallel Parsing (XML)

Selected Projects

- Jan 2019 – **Latent Space Optimization Framework for Conditional Generative Models.**
Dec 2019 Ongoing Research Project, The current work is within optimization aspects of conditional generative networks towards the task of controlled image synthesis.
- Sep 2019 – **Pytorch Implementation of Dense Captioning.**
Dec 2019 Ongoing Open Source Project, Pytorch implementation of Dense Captioning with Joint Inference and Visual Context trained on the Visual Genome dataset.
- Sep 2019 – **Introduction to Computer Vision Course Project.**
Dec 2019 Project: Pose Tracking, The course led me to implement few algorithms from scratch using Numpy only. I implemented Morphological Operations, Hough Transformation, Histogram based Equalizaion and Segmentation, Ligteneing Correction, Canny Edge Detection and Template based Image Tracking.
- April 2019 – **Experiments with Actor Critique Methods for Deep Reinforcement Learning.**
June 2019 I performed a survey on existing approaches for a particular OpenAI gym environment, Bipedal-Walker-V2. Further, I found evidence of actor-critic methods dominating the baseline with a significant benchmark. My current implementation is built on top of the Asynchronous Actor Critique (A3C) Algorithm.

- April 2019 – **Seminar In Statistical Language Modeling Course Project.**
- June 2019 The course led me to read and summarize 40 selective papers that are popular contributed to Statistical language modeling. For the course project, I read and analyzed text to image model and built one from scratch. This course project also gave me a better understanding of the attention mechanism and the behind the scenes workings of transformers.
- Sep 2018 – **Google Analytics Revenue Prediction.**
- Dec 2018 This project is my contribution to a live Kaggle competition by Google. Google gave the task to predict the total revenue generated per customer based on the customer dataset of a Google Analytics Merchandise Store. In short, I have tackled a regression task
- June 2017 – **S-LSTM-GAN: Shared Recurrent Neural Networks with Adversarial Training.**
- April 2018 This project introduces a shared layer generative model. Recurrent networks are used to work with continuous data sequences, in this work, I combined them with the generative modeling mechanism and adversarial training to evaluate their receptive field towards the task of image generation.
- Dec 2016 – **Experiments with Generative Adversarial Networks.**
- Dec 2017 Experimented on the GAN framework to evaluate the performance of various image compression and image super-resolution techniques. I also worked with adversarial examples, specifically experimenting with generating noise that affects single image classification.

Selected Publications

- ICDECT - *S-LSTM-GAN: Shared Recurrent Neural Networks with Adversarial Training.* **A Adate**, BK Tripathy - International Conference on Data Engineering and Communication Technology, 2017 [\[Link\]](#)
- ICIIT - *Understanding How Adversarial Noise Affects Single Image Classification.* **A Adate**, R Saxena, Don.S - International Conference on Intelligent Information Technologies, 2017 [\[Link\]](#)
- SOCPROS - *Understanding Single Image Super-Resolution Techniques with Generative Adversarial Networks.* **A Adate**, BK Tripathy - Soft Computing for Problem Solving, 2018 [\[Link\]](#)

Technical Skills

Programming

Python3, Matlab, C , C++ , LaTeX, Octave

Frameworks / Libraries

Pytorch, scikit-learn, OpenCV

Tensorflow, Keras , Nvidia CUDA - cuFFT and cuDNN

Selected Coursework

- Graduate Machine Learning, Deep Learning Foundations From Scratch
Deep Reinforcement Learning From Scratch, Natural Language Processing
Seminar in Statistical language Modeling, Statistics for Life Sciences
Frameworks for Artificial Intelligence, Data Science Seminar
- Undergraduate Soft Computing, Image and Vision Computing, Data Structures
Discrete Mathematical Structures, Linear Algebra, Statistics and Reliability, Graph Theory
- External Machine Learning (Coursera), Nerual Networks for Machine Learning (Coursera),
Data-Structures Specialization (Coursera)

References

Available upon request